

Problem 2

The intensity of the gravitational field at a certain point is $G = 9.8 \text{ N/kg}$, and the gravitational potential is $\varphi = -67 \text{ MJ/kg}$. If a body with mass $m = 2 \text{ kg}$ is placed at this point:

- (a) What is the gravitational force acting on the body?
- (b) What gravitational potential energy will the body acquire?

Solution

- (a) The gravitational force acting on the body is given by:

$$F_g = mG$$

Substitute the given values:

$$F_g = (2 \text{ kg}) \times (9.8 \text{ N/kg}) = 19.6 \text{ N}$$

- (b) The gravitational potential energy is given by:

$$U_g = m\varphi$$

Convert the potential to SI units:

$$\varphi = -67 \text{ MJ/kg} = -67 \times 10^6 \text{ J/kg}$$

Then calculate the potential energy:

$$U_g = (2 \text{ kg}) \times (-67 \times 10^6 \text{ J/kg})$$

$$U_g = -1.34 \times 10^8 \text{ J}$$

Conclusion:

- The gravitational force on the body is 19.6 N. - The gravitational potential energy of the body is $-1.34 \times 10^8 \text{ J}$.